

Name: Kukua Eshun
ID: 90782028
Cohort C

```
MariaDB [(none)]> create database Project;  
Query OK, 1 row affected (0.005 sec)
```

```
MariaDB [(none)]> use Project;  
Database changed
```

```
MariaDB [Project]> create table Department(deptNo varchar(5)  
primary key,deptName varchar(25),mgrEmpNo varchar(5) unique);  
Query OK, 0 rows affected (0.051 sec)
```

```
MariaDB [Project]> describe department;
```

Field	Type	Null	Key	Default	Extra
deptNo	varchar(5)	NO	PRI	NULL	
deptName	varchar(25)	YES		NULL	
mgrEmpNo	varchar(5)	YES	UNI	NULL	

```
3 rows in set (0.011 sec)
```

```
MariaDB [Project]> insert into Department values  
-> ('D1','IT',NULL),  
-> ('D2','HR',NULL),  
-> ('D3','Finance',NULL),  
-> ('D4','Marketing',NULL),  
-> ('D5','Operations',NULL),  
-> ('D6','Sales',NULL),  
-> ('D7','R&D',NULL),  
-> ('D8','Support',NULL),  
-> ('D9','Logistics',NULL),  
-> ('D10','Admin',NULL);
```

```
Query OK, 10 rows affected (0.003 sec)
```

```
MariaDB [Project]> create table Employee(empNo varchar(5)  
primary key,fName varchar(20),lName varchar(20),address  
varchar(30),DOB date,sex char(1),position varchar(15),deptNo  
varchar(5),foreign key(deptNo) references  
Department(deptNo),constraint chk_sex check(sex in  
('M','F')),constraint chk_position check(position in  
('Manager','Team Leader','Analyst','Software Developer')));  
Query OK, 0 rows affected (0.038 sec)
```

```
MariaDB [Project]> describe employee;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(5)	NO	PRI	NULL	
fName	varchar(20)	YES		NULL	
lName	varchar(20)	YES		NULL	
address	varchar(30)	YES		NULL	
DOB	date	YES		NULL	
sex	char(1)	YES		NULL	
position	varchar(30)	YES		NULL	
deptNo	varchar(5)	YES	MUL	NULL	

```
8 rows in set (0.004 sec)
```

```
MariaDB [Project]> insert into Employee values
```

```
-> ('E1','John','Doe','Accra','2001-05-10','M','Manager','D1'),
-> ('E2','Ama','Mensah','Tema','2000-03-12','F','Team
Leader','D1'),
-> ('E3','Kofi','Boateng','Kumasi','1999-07-22','M','Analyst','D2
'),
-> ('E4','Abena','Owusu','Cape
Coast','2002-01-15','F','Software Developer','D2'),
-> ('E5','Yaw','Darko','Accra','1998-11-09','M','Analyst','D3'),
-> ('E6','Akosua','Bonsu','Takoradi','2001-09-30','F','Team
Leader','D3'),
-> ('E7','Kojo','Mensah','Tamale','1997-06-18','M','Software
Developer','D4'),
-> ('E8','Efua','Asare','Ho','2003-04-25','F','Manager','D4'),
-> ('E9','Kwame','Osei','Accra','1996-12-05','M','Analyst','D5'),
-> ('E10','Adjoa','Nkrumah','Koforidua','2000-08-14','F','Softwar
e Developer','D5');
```

```
MariaDB [Project]> create table Project(projNo varchar(5)
primary key,projName varchar(25),deptNo varchar(5),foreign
key(deptNo) references Department(deptNo));
Query OK, 0 rows affected (0.033 sec)
```

```
MariaDB [Project]> describe project;
```

Field	Type	Null	Key	Default	Extra
projNo	varchar(5)	NO	PRI	NULL	
projName	varchar(25)	YES		NULL	
deptNo	varchar(5)	YES	MUL	NULL	

3 rows in set (0.003 sec)

```
MariaDB [Project]> insert into Project values
-> ('P1','AI System','D1'),
-> ('P2','Payroll','D2'),
-> ('P3','Banking App','D3'),
-> ('P4','E-Commerce','D4'),
-> ('P5','Logistics Tracker','D5'),
-> ('P6','CRM System','D6'),
-> ('P7','Research Tool','D7'),
-> ('P8','Helpdesk','D8'),
-> ('P9','Fleet System','D9'),
-> ('P10','Admin Portal','D10');
```

Query OK, 10 rows affected (0.001 sec)

```
MariaDB [Project]> create table WorksOn(empNo
varchar(5),projNo varchar(5),dateWorked date,hoursWorked
int,primary key(empNo,projNo,dateWorked),foreign key(empNo)
references Employee(empNo),foreign key(projNo) references
Project(projNo),constraint chk_hours check(hoursWorked between
0 and 40));
```

Query OK, 0 rows affected (0.030 sec)

```
MariaDB [Project]> describe workson;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(5)	NO	PRI	NULL	
projNo	varchar(5)	NO	PRI	NULL	
dateWorked	date	NO	PRI	NULL	
hoursWorked	int(11)	YES		NULL	

4 rows in set (0.005 sec)

```
MariaDB [Project]> insert into WorksOn values
-> ('E1','P1','2026-06-01',5),
-> ('E2','P1','2026-06-01',6),
-> ('E3','P2','2026-06-02',7),
-> ('E4','P2','2026-06-02',8),
-> ('E5','P3','2026-06-03',4),
-> ('E6','P3','2026-06-03',6),
```

```

-> ('E7','P4','2026-06-04',5),
-> ('E8','P4','2026-06-04',3),
-> ('E9','P5','2026-06-05',6),
-> ('E10','P5','2026-06-05',7);
Query OK, 10 rows affected (0.002 sec)

```

```

MariaDB [Project]> create view FemaleProjects as select
p.projNo,p.projName from Project p join Department d on
p.deptNo=d.deptNo join Employee e on d.mgrEmpNo=e.empNo where
e.sex='F' order by p.projNo;
Query OK, 0 rows affected (0.030 sec)

```

```

MariaDB [Project]> describe femaleprojects;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| projNo     | varchar(5)    | NO   |     | NULL    |       |
| projName   | varchar(25)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.009 sec)

```

```

MariaDB [Project]> create view TableAttributes as select
e.empNo, e.fName,e.lName,p.projName,w.hoursWorked from
Employee as e join WorksOn as w on e.empNo=w.empNo join
Project as p on w.projNo=p.projNo;
Query OK, 0 rows affected (0.021 sec)

```

```

MariaDB [Project]> create view
EmpProject(empNo,projNo,totalHours) as select
w.empNo,w.projNo,sum(w.hoursWorked) from WorksOn w group by
w.empNo,w.projNo;
Query OK, 0 rows affected (0.023 sec)

```

```

MariaDB [Project]> select * from EmpProject;
+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
| E1     | P1     | 5          |
| E10    | P5     | 7          |
| E2     | P1     | 6          |
| E3     | P2     | 7          |
| E4     | P2     | 8          |
| E5     | P3     | 4          |
+-----+-----+-----+

```

E6	P3	6
E7	P4	5
E8	P4	3
E9	P5	6

10 rows in set (0.001 sec)

MariaDB [case2]> SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';
 Empty set (0.001 sec)

MariaDB [Project]> select count(projNo) from EmpProject where empNo='E1';

count(projNo)
1

1 row in set (0.001 sec)

MariaDB [Project]> select empNo,totalHours from EmpProject group by empNo;

empNo	totalHours
E1	5
E10	7
E2	6
E3	7
E4	8
E5	4
E6	6
E7	5
E8	3
E9	6

10 rows in set (0.001 sec)

